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Aturan Mason
Bentuk Keterkontrolan

MA4171 Teori Kontrol Linear

Mason's Rule

- Mason's gain formula (MGF) is a method for finding the transfer function of a linear signal-flow graph (SFG).
- The formula was derived by Samuel Jefferson Mason, whom it is also named after. MGF is an alternate method to finding the transfer function algebraically by labeling each signal, writing down the equation for how that signal depends on other signals, and then solving the multiple equations for the output signal in terms of the input signal.
- MGF provides a step by step method to obtain the transfer function from a SFG. Often, MGF can be determined by inspection of the SFG. The method can easily handle SFGs with many variables and loops including loops with inner loops.
- MGF comes up often in the context of control systems, microwave circuits and digital filters because these are often represented by SFGs.

Mason's Rule

- Fungsi transfer $G(s)$ dari suatu system, direpresentasikan melalui grafik aliran sinyal

$$G(s) = \frac{R(s)}{Q(s)} = \frac{\sum_k F_k \Delta_k}{\Delta}$$

dengan

F_k = gain dari aliran langsung ke- k yang menghubungkan input ke output

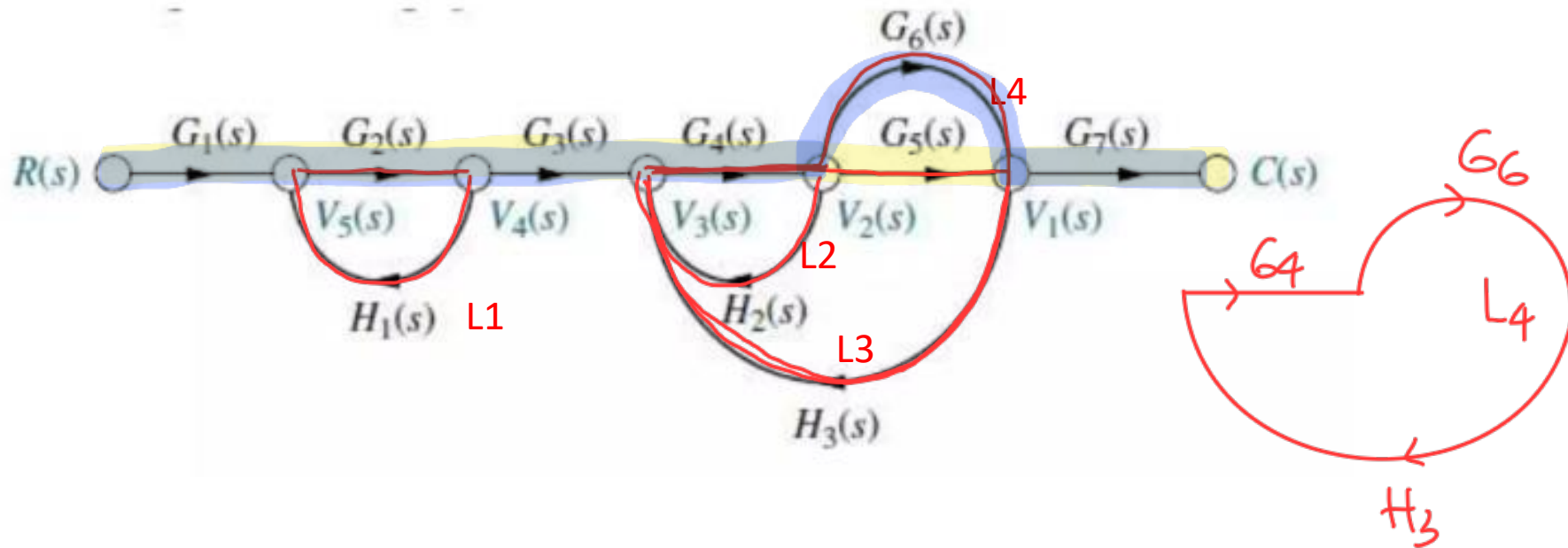
$\Delta_k = 1 - \sum$ gain loop non touching sepanjang aliran langsung

$\Delta = 1 - \sum$ gain dari setiap loop + $\sum$ gain dari kombinasi 2 loop non touching - $\sum$ gain dari kombinasi 3 loop non touching + ...
Definisikan, gain loop L_i

Perhatikan, $\Delta = 1 - \sum L_i + \sum L_i L_j - \sum L_i L_j L_k + \dots$

CONTOH

- Perhatikan signal flow graph berikut. Tentukan fungsi transfer.



Perhatikan aliran signal tersebut, maka diperoleh :

- Dua buah aliran langsung (Forward) F_1, F_2
- Empat buah single loop L_1, L_2, L_3, L_4
- Tiga buah kombinasi 2 loop yang non-touching L_1L_2, L_1L_3, L_1L_4

Single loop yang diperoleh adalah

$$L1. G_2(s)H_1(s)$$

$$L2. G_4(s)H_2(s)$$

$$L3. G_4(s)G_5(s)H_3(s)$$

$$L4. G_4(s)G_6(s)H_3(s)$$

Aliran langsung (forward path) adalah

$$F1. G_1(s)G_2(s)G_3(s)G_4(s)G_5(s)G_7(s)$$

$$F2. G_1(s)G_2(s)G_3(s)G_4(s)G_6(s)G_7(s)$$

Perhatikan bahwa, semua aliran langsung menyentuh (node) setiap Loop, maka $\Delta_1 = 1$ dan $\Delta_2 = 1$

Kombinasi 2 loop non-touching

$$NT1. [G_2(s)H_1(s)][G_4(s)H_2(s)] \quad L1L2$$

$$NT2. [G_2(s)H_1(s)][G_4(s)G_5(s)H_3(s)] \quad L1L3$$

$$NT3. [G_2(s)H_1(s)][G_4(s)G_6(s)H_3(s)] \quad L1L4$$

Kofaktor system

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + (L_1L_2 + L_1L_3 + L_1L_4)$$

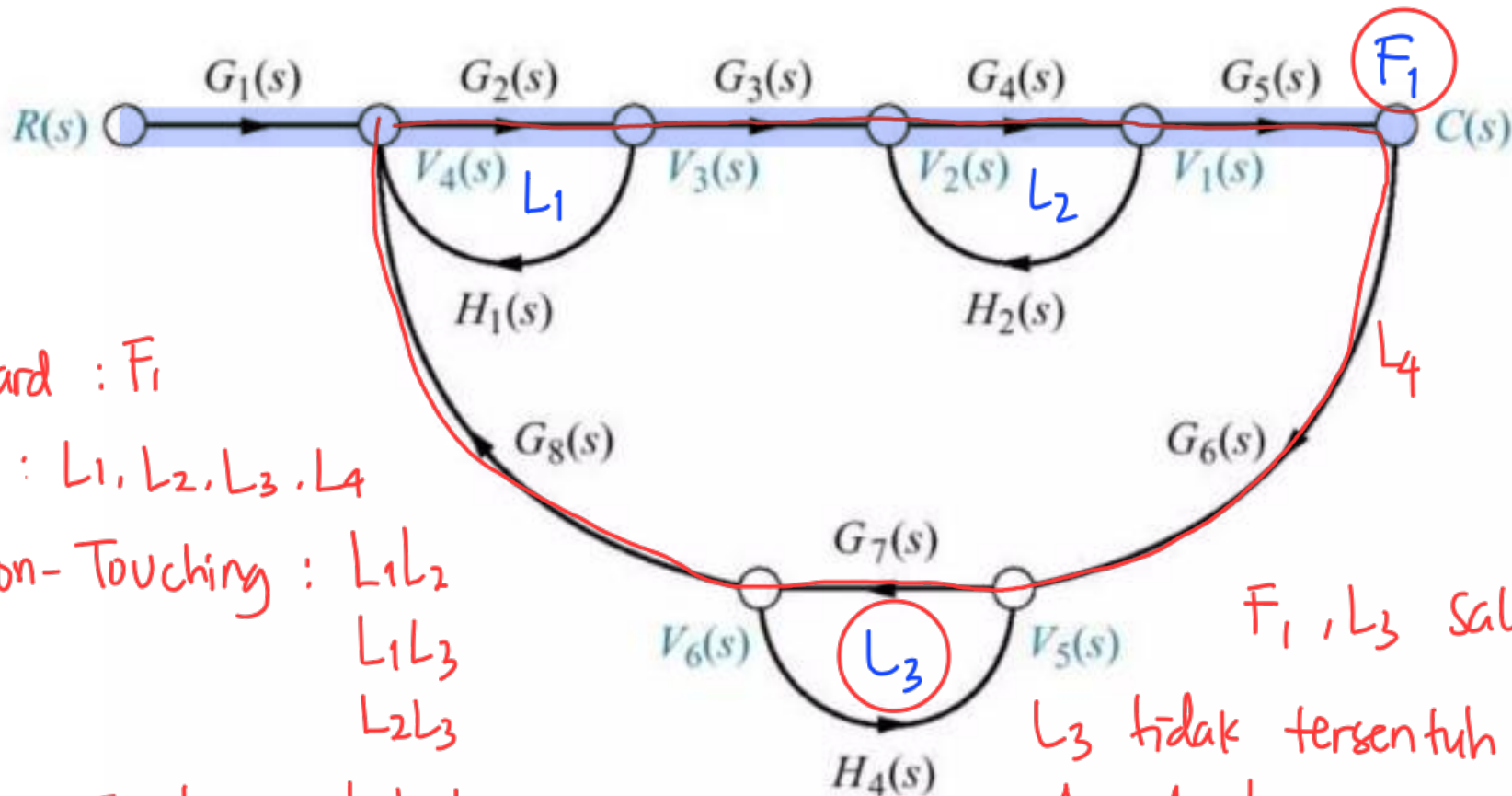
Fungsi Transfer

$$G(s) = \frac{F_1\Delta_1 + F_2\Delta_2}{\Delta}$$

$$\Delta = 1 - [G_2H_1 + G_4H_2 + G_4G_5H_3 + G_4G_6H_3] + [G_2G_4H_1H_2 + G_2G_4G_5H_1H_3 + G_2G_4G_6H_1H_3]$$

Soal Latihan

- Perhatikan signal flow graph berikut. Tentukan fungsi transfer.



Forward : F_1

Loop : L_1, L_2, L_3, L_4

2 Non-Touching : $L_1 L_2$

$L_1 L_3$

$L_2 L_3$

3 Non-Touching : $L_1 L_2 L_3$

F_1, L_3 saling bebas

L_3 tidak tersentuh F_1

$$\Delta_1 = 1 - L_3$$

Perhatikan aliran signal tersebut, maka diperoleh :

- Satu buah aliran langsung (Forward)
- Empat buah single loop
- Tiga buah kombinasi 2 loop yang non-touching
- Satu kombinasi 3 loop non-touching

$$F1. \quad G_1(s)G_2(s)G_3(s)G_4(s)G_5(s)$$

$$L1. \quad G_2(s)H_1(s) \quad L3. \quad G_7(s)H_4(s)$$

$$L2. \quad G_4(s)H_2(s) \quad L4. \quad G_2(s)G_3(s)G_4(s)G_5(s)G_6(s)G_7(s)G_8(s)$$

$$L1 \text{ and } L2: \quad G_2(s)H_1(s)G_4(s)H_2(s) \quad L2 \text{ and } L3: \quad G_4(s)H_2(s)G_7(s)H_4(s)$$

$$L1 \text{ and } L3: \quad G_2(s)H_1(s)G_7(s)H_4(s)$$

$$L1, L2, L3: \quad G_2(s)H_1(s)G_4(s)H_2(s)G_7(s)H_4(s)$$

$$\Delta = 1 - \sum L_i + \sum L_i L_j - \sum L_i L_j L_k$$

$$\begin{aligned} \Delta = 1 &- [G_2(s)H_1(s) + G_4(s)H_2(s) \\ &+ G_7(s)H_4(s) + G_2(s)G_3(s)G_4(s)G_5(s)G_6(s)G_7(s)G_8(s)] \\ &+ [G_2(s)H_1(s)G_4(s)H_2(s) + G_2(s)H_1(s)G_7(s)H_4(s) \\ &+ G_4(s)H_2(s)G_7(s)H_4(s)] \\ &- [G_2(s)H_1(s)G_4(s)H_2(s)G_7(s)H_4(s)] \end{aligned}$$

$$\Delta_1 = 1 - L_3$$

$$\Delta_1 = 1 - G_7(s)H_4(s)$$

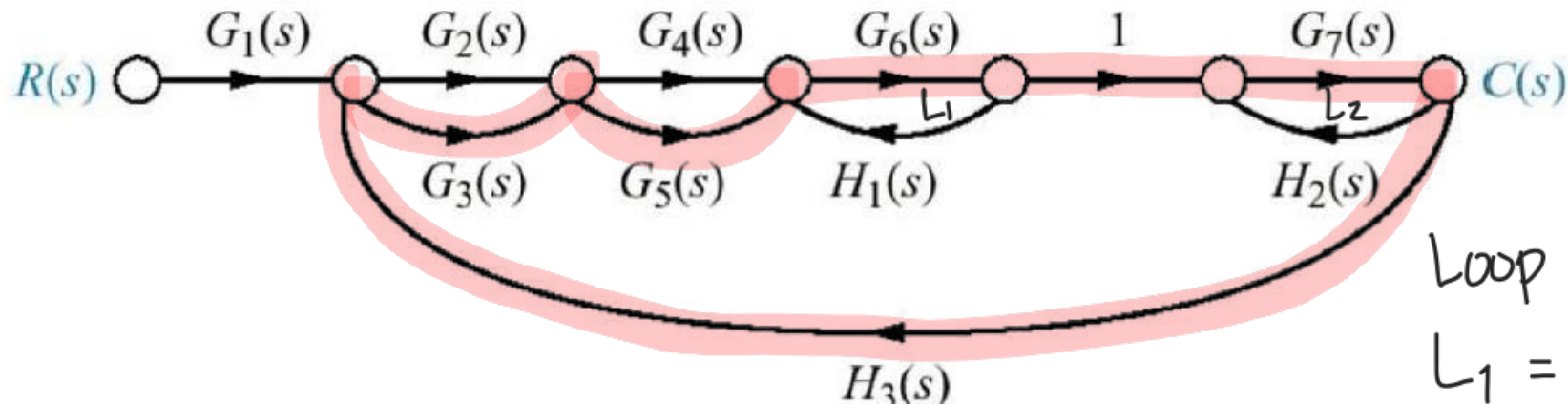
Aliran langsung tidak menyentuk loop L_4

Fungsi Transfer

$$G(s) = \frac{F_1 \Delta_1}{\Delta} = \frac{G_1 G_2 G_3 G_4 G_5 (1 - G_7 H_4)}{\Delta}$$

Soal Latihan

- Perhatikan signal flow graph berikut. Dengan menggunakan aturan Mason tentukan fungsi transfer.



Grafik Aliran Sinyal :

- Empat Forward Path
- Enam Loop
- 1 kombinasi 2 loop Nontouching : $L_1 L_2$

Loop :

$$L_1 = G_6 H_1 \quad L_2 = G_7 H_2$$

$$L_3 = G_2 G_4 G_6 G_7 H_3$$

$$L_4 = G_2 G_5 G_6 G_7 H_3$$

$$L_5 = G_3 G_4 G_6 G_7 H_3$$

$$L_6 = G_3 G_5 G_6 G_7 H_3$$

Forward Path :

$$F_1 = G_1 G_2 G_4 G_6 G_7$$

$$F_2 = G_1 G_3 G_4 G_6 G_7$$

$$F_3 = G_1 G_2 G_5 G_6 G_7$$

$$F_4 = G_1 G_3 G_5 G_6 G_7$$

$$\Delta_1 = 1$$

$$\Delta_2 = 1$$

$$\Delta_3 = 1$$

$$\Delta_4 = 1$$

Semua forward
menyentuh loop

$$\begin{aligned} \text{Kofaktor : } \Delta &= 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6) + (L_1 L_2) \\ &= 1 - \dots \end{aligned}$$

$$\text{Fungsi Transfer : } G(s) = \frac{F_1 \Delta_1 + F_2 \Delta_2 + F_3 \Delta_3 + F_4 \Delta_4}{\Delta}$$

Persamaan Ruang Keadaan untuk Fungsi Transfer

- ✓ Bentuk kanonik keterkontrolan
- ✓ Bentuk kanonik keterobservasian
- ✓ Bentuk kanonik diagonal
- ✓ Bentuk kanonik Jordan
- Signal-flow-graph untuk persamaan ruang keadaan

$$\frac{Y(s)}{U(s)} = C(SI - A)^{-1}B + D //$$

SISO

$$\begin{aligned}\dot{\vec{x}} &= A\vec{x} + Bu \\ y &= C\vec{x} + Du\end{aligned}$$

Fungsi Transfer :

$$\mathcal{L}\{\dot{\vec{x}}\} = \mathcal{L}\{A\vec{x} + Bu\}$$

$$sX(s) = AX(s) + BU(s)$$

$$sX(s) - AX(s) = BU(s)$$

$$X(s) [sI - A] = BU(s)$$

$$X(s) = [sI - A]^{-1}BU(s)$$

$$\mathcal{L}\{y\} = \mathcal{L}\{C\vec{x} + Du\}$$

$$Y(s) = CX(s) + DU(s)$$

$$Y(s) = C[sI - A]^{-1}BU(s) + DU(s)$$

invers trans
Laplace

Bentuk Kanonik Keterkontrolan

Perhatikan model sistem linear orde- n

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} u' + b_n u.$$

Persamaan diatas dapat juga ditulis dalam bentuk:

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

Sistem ini termasuk SISO.

Persamaan ruang keadaan dalam bentuk kanonik keterkontrolan dari system diatas dapat ditulis dalam bentuk:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \cdot \\ \cdot \\ \cdot \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ \cdot \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [b_n - a_n b_0 \mid b_{n-1} - a_{n-1} b_0 \mid \cdots \mid b_1 - a_1 b_0] \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} + b_0 u$$

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

$$= b_0 + \frac{(b_1 - a_1 b_0) s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0) s + (b_n - a_n b_0)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

$$Y(s) = b_0 U(s) + \frac{(b_1 - a_1 b_0) s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0) s + (b_n - a_n b_0)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} U(s)$$

$$Y(s) = b_0 U(s) + \hat{Y}(s)$$

$$\hat{Y}(s) = \frac{(b_1 - a_1 b_0) s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0) s + (b_n - a_n b_0)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} U(s)$$

$$\frac{\hat{Y}(s)}{(b_1 - a_1 b_0) s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0) s + (b_n - a_n b_0)}$$

$$= \frac{U(s)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} = Q(s)$$

$$U(s) = s^n Q(s) + a_1 s^{n-1} Q(s) + \dots + a_{n-1} s Q(s) + a_n Q(s)$$

$$s^n Q(s) = -a_1 s^{n-1} Q(s) - \dots - a_{n-1} s Q(s) - a_n Q(s) + U(s) \leftarrow$$

$$\hat{Y}(s) = (b_1 - a_1 b_0) s^{n-1} Q(s) + \dots + (b_{n-1} - a_{n-1} b_0) s Q(s) + (b_n - a_n b_0) Q(s)$$

Dengan memisalkan :

$$X_1(s) = Q(s)$$

$$X_2(s) = s Q(s)$$

$$X_3(s) = s^2 Q(s)$$

$\vdots$

$$X_{n-1}(s) = s^{n-2} Q(s)$$

$$X_n(s) = s^{n-1} Q(s)$$

$$s X_1(s) = s Q(s) = X_2(s)$$

$$s X_2(s) = s^2 Q(s) = X_3(s)$$

$$s X_3(s) = s^3 Q(s) = X_4(s)$$

$\vdots$

$$s X_{n-1}(s) = s^{n-1} Q(s) = X_n(s)$$

$$s X_n(s) = s^n Q(s)$$

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = x_3(t)$$

$$\dot{x}_3(t) = x_4(t)$$

$\vdots$

$$\dot{x}_{n-1}(t) = x_n(t)$$

$$\dot{x}_n(t) = \dots$$

Langkah-langkah untuk menunjukkan persamaan tersebut adalah sbb:

Dengan membagi langsung persamaan sebelumnya dapat dirubah menjadi:

$$\frac{Y(s)}{U(s)} = b_0 + \frac{(b_1 - a_1 b_0)s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0)s + (b_n - a_n b_0)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

atau $Y(s) = b_0 U(s) + \hat{Y}(s)$

dengan $\hat{Y}(s) = \frac{(b_1 - a_1 b_0)s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0)s + (b_n - a_n b_0)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} U(s)$

atau
$$\frac{\hat{Y}(s)}{(b_1 - a_1 b_0)s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0)s + (b_n - a_n b_0)}$$
$$= \frac{U(s)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} = Q(s)$$

Dari persamaan terakhir, diperoleh persamaan berikut:

$$\boxed{s^n Q(s)} = -a_1 s^{n-1} Q(s) - \cdots - a_{n-1} s Q(s) - a_n Q(s) + U(s)$$

$$\hat{Y}(s) = (b_1 - a_1 b_0) s^{n-1} Q(s) + \cdots + (b_{n-1} - a_{n-1} b_0) s Q(s) + (b_n - a_n b_0) Q(s)$$

Dengan memisalkan:

$X_1(s) = Q(s)$		$sX_1(s) = X_2(s)$		$\dot{x}_1 = x_2$
$X_2(s) = sQ(s)$		$sX_2(s) = X_3(s)$		$\dot{x}_2 = x_3$
$\vdots$		$\vdots$		$\vdots$
$\vdots$		$\vdots$		$\vdots$
$\vdots$		$\vdots$		$\vdots$
$X_{n-1}(s) = s^{n-2} Q(s)$		$sX_{n-1}(s) = X_n(s)$		$\dot{x}_{n-1} = x_n$
$X_n(s) = s^{n-1} Q(s)$		$sX_n = \boxed{s^n Q(s)}$		

Dengan demikian persamaan ruang keadaan dalam bentuk kanonik keterkontrolan diperoleh.

Bentuk kanonik keterobservasian

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \cdot \\ \cdot \\ \cdot \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ \cdot & \cdot & & \cdot & \cdot \\ \cdot & \cdot & & \cdot & \cdot \\ \cdot & \cdot & & \cdot & \cdot \\ 0 & 0 & \cdots & 1 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} + \begin{bmatrix} b_n - a_n b_0 \\ b_{n-1} - a_{n-1} b_0 \\ \cdot \\ \cdot \\ \cdot \\ b_1 - a_1 b_0 \end{bmatrix} u$$

$$y = [0 \quad 0 \quad \cdots \quad 0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_{n-1} \\ x_n \end{bmatrix} + b_0 u$$

Contoh

- Tentukan persamaan ruang keadaan dalam bentuk kanonik keterkontrolan dari fungsi transfer berikut:

$$(a) \frac{Y(s)}{U(s)} = \frac{s+3}{s^2+3s+2}$$

$$(b) \frac{Y(s)}{U(s)} = \frac{s+1}{s^2+2s+3}$$

$$(c) \frac{Y(s)}{U(s)} = \frac{10.4s^2+47s+160}{s^3+14s^2+56s+160}$$

(a) Diberikan fungsi transfer :

$$\frac{Y(s)}{U(s)} = \frac{s+3}{s^2+3s+2} \quad \text{system kontrol linear orde-2}$$

output

Perhatikan, $\boxed{\frac{Y(s)}{s+3}} = \boxed{\frac{U(s)}{s^2+3s+2}} = Q(s)$

input

$$\bullet \frac{U(s)}{s^2+3s+2} = Q(s) \Rightarrow U(s) = s^2 Q(s) + 3s Q(s) + 2Q(s)$$

$$\boxed{s^2 Q(s) = -3s Q(s) - 2Q(s) + U(s)}$$

Definisikan : $X_1(s) = Q(s)$
 $X_2(s) = sQ(s) \Rightarrow$

$$\begin{aligned} sX_1(s) &= sQ(s) = X_2(s) \\ sX_2(s) &= s^2 Q(s) = -3s Q(s) - 2Q(s) + U(s) \\ &= -2X_1(s) - 3X_2(s) + U(s) \end{aligned}$$

Invers Transf. Laplace :

$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -2x_1(t) - 3x_2(t) + u(t) \end{aligned}$$

$$\bullet \frac{Y(s)}{s+3} = Q(s) \Rightarrow Y(s) = sQ(s) + 3Q(s)$$
$$Y(s) = 3X_1(s) + X_2(s)$$

Invers Transf. Laplace : $y(t) = 3x_1(t) + x_2(t)$

Pers. Ruang Keadaan : Kanonik Keterkontrolan

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Perhatikan, $\frac{Y(s)}{U(s)} = \frac{s+3}{s^2+3s+2}$ yang akan dibentuk menjadi system control linear orde 2

Maka buatlah perbandingan sedemikian sehingga :

$$\text{output} \boxed{\frac{Y(s)}{s+3}} = \boxed{\frac{U(s)}{s^2+3s+2}} = Q(s)$$

state

Perhatikan

$$\frac{U(s)}{s^2+3s+2} = Q(s) \Rightarrow U(s) = s^2Q(s) + 3sQ(s) + 2Q(s)$$
$$s^2Q(s) = -2Q(s) - 3sQ(s) + U(s)$$

Definisikan : $X_1(s) = Q(s)$ dan $X_2(s) = sQ(s)$, maka diperoleh

$$sX_1(s) = sQ(s) = X_2(s)$$
$$sX_2(s) = s^2Q(s) = -2Q(s) - 3sQ(s) + U(s) = -2X_1(s) - 3X_2(s) + U(s)$$

Dengan menggunakan invers Laplace, diperoleh state :

$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = -2x_1(t) - 3x_2(t) + u(t)$$

Perhatikan

$$\frac{Y(s)}{s+3} = Q(s) \Rightarrow Y(s) = sQ(s) + 3Q(s) = 3X_1(s) + X_2(s)$$

Dengan menggunakan invers Laplace, diperoleh output :

$$y(t) = 3x_1(t) + x_2(t)$$

Dengan demikian, ruang keadaannya adalah

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$$
$$y = [3 \quad 1] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Perhatikan, $\frac{Y(s)}{U(s)} = \frac{10.4s^2+47s+160}{s^3+14s^2+56s+160}$ yang akan dibentuk menjadi system control linear orde 2

Maka buatlah perbandingan sedemikian sehingga :

$$\text{output } \boxed{\frac{Y(s)}{10.4s^2 + 47s + 160}} = \boxed{\frac{U(s)}{s^3 + 14s^2 + 56s + 160}} = Q(s)$$

state

Perhatikan

$$\frac{U(s)}{s^3 + 14s^2 + 56s + 160} = Q(s) \Rightarrow U(s) = s^3 Q(s) + 14s^2 Q(s) + 56s Q(s) + 160 Q(s)$$
$$s^3 Q(s) = -160 Q(s) - 56s Q(s) - 14s^2 Q(s) + U(s)$$

Definisikan : $X_1(s) = Q(s)$, $X_2(s) = sQ(s)$ dan $X_3(s) = s^2 Q(s)$ maka diperoleh

$$sX_1(s) = sQ(s) = X_2(s)$$

$$sX_2(s) = s^2 Q(s) = X_3(s)$$

$$sX_3(s) = s^3 Q(s) = -160Q(s) - 56sQ(s) - 14s^2 Q(s) + U(s)$$
$$= -160X_1(s) - 56X_2(s) - 14X_3(s) + U(s)$$

Dengan menggunakan invers Laplace, diperoleh state :

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = x_3(t)$$

$$\dot{x}_3(t) = -160x_1(t) - 56x_2(t) - 14x_3(t) + u(t)$$

Perhatikan

$$\frac{Y(s)}{10.4s^2 + 47s + 160} = Q(s) \Rightarrow Y(s) = 160Q(s) + 47sQ(s) + 10.4s^2Q(s)$$
$$Y(s) = 160X_1(s) + 47X_2(s) + 10.4X_3(s)$$

Dengan menggunakan invers Laplace, diperoleh output :

$$y(t) = 160x_1(t) + 47x_2(t) + 10.4x_3(t)$$

Dengan demikian, ruang keadaannya adalah

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -160 & -56 & -14 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u$$
$$y = [160 \quad 47 \quad 10.4] \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$